Pose-Based Human State Recognition for Search-and-Rescue Drones

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Abstract

We build and study a pipeline for person detection, tracking, pose estimation, and state classification in aerial search-and-rescue (SAR) video, and treat the pipeline as a controlled investigation of four research questions. First, we quantify the degradation of a ground-trained two-dimensional pose estimator on aerial imagery and show that it is governed by person size rather than by viewpoint: on 22,319 UAV-Human persons, the percentage of correct keypoints at a threshold of 0.1 falls from 0.944 for people taller than one hundred pixels to 0.440 in the fifty-to-one-hundred-pixel band and 0.091 below fifty pixels. Second, we ask whether an explicit pose representation outperforms an appearance-only baseline for triage-state classification; under a strict video-level split, the appearance model narrowly leads (macro-averaged F1 of 0.399 against 0.363), and both models fail on the rare motionless class. Third, we evaluate frequency-domain restoration of degraded aerial frames and demonstrate that improvements in pixel quality do not imply improvements in task performance: on additive noise, the non-local means filter raises the peak signal-to-noise ratio yet lowers detection average precision below the degraded baseline, whereas a simple median filter improves it. Fourth, we compare the SIFT descriptor, the ORB descriptor, and a compact self-supervised descriptor on the HPatches benchmark, and apply a homography-based registration that converts the per-frame triage overlay into a single victim map. All final results derive from one fine-tuned detector checkpoint that attains an average precision at an intersection-over-union of 0.5 of 0.709 on the VisDrone person subset.

1. Introduction and Motivation

In a search-and-rescue (SAR) operation, an unmanned aerial vehicle can sweep a disaster area far more quickly than a ground team. Nevertheless, the raw aerial video still requires a human operator to locate people and to judge their condition, and this human step is the bottleneck of the process. A system that merely detects people produces a map of dots: it tells the operator where people are but not who needs help first. The information that matters for triage is the *state* of each person — whether a person is lying motionless, and therefore possibly unconscious or injured, or standing and moving, and therefore likely able to assist themselves. This distinction between a motionless person and an active one is fundamentally a question of human posture, and posture is estimated by human pose estimation. This observation motivates the entire project.

Modern two-dimensional pose estimators, however, are trained predominantly on ground-level photographs, in which people are large and viewed from the side or front. A drone observes people from above, at steep angles, and at very small pixel sizes. The central practical risk is therefore a *domain gap*: a pose estimator that is accurate on the ground may be unreliable in the air. This work constructs the complete aerial SAR pipeline and uses it as an instrument to answer two research questions from the original project proposal, together with two additional questions that broaden the computer-vision scope of the project.

- **RQ1 (domain gap).** How much does a two-dimensional pose estimator trained on ground-level images degrade under aerial viewpoints and small person scales, and how much of that loss can lightweight fine-tuning recover?

- **RQ2 (does pose help?).** For classifying a person's state from a drone, does an explicit pose representation outperform an appearance-only baseline?
- **RQ3 (restoration).** Does classical, frequency-domain deblurring and denoising of aerial frames recover not only pixel quality but also the downstream detection and pose accuracy?
- **RQ4 (matching and mapping).** How do the SIFT descriptor, the ORB descriptor, and a learned descriptor compare under viewpoint, scale, and illumination change, and can a homography-based registration aggregate the per-frame triage into a single map?

The two added questions follow directly from the deployment setting. Drones vibrate and operate in poor light, which produces blurred and noisy frames and motivates restoration (RQ3). Operators need a single map of the searched area rather than a stream of per-frame images, which motivates registration (RQ4). Together, the four questions give a realistic account of where an aerial triage pipeline succeeds and where it fails.

The contribution of the project is therefore not a single new model but a controlled, end-to-end account of the aerial triage problem, in which every claim is supported by a full-scale measurement produced by our own evaluators. Each research question is designed so that its answer is informative whether it is positive or negative: a negative result, such as the finding that an explicit pose representation does not yet outperform an appearance baseline at aerial scale, is reported and analysed rather than concealed, because it locates the true difficulty of the problem.

2. Related Work

Aerial person detection. Object detection from aerial platforms has been driven by the VisDrone benchmark [1], which provides annotated drone imagery across a range of altitudes and viewpoints. Single-stage detectors of the YOLO family are the standard practical choice for real-time aerial detection; we use YOLO11 [2] as our detector. Multi-object tracking associates detections across frames; we use ByteTrack [3], which associates every detection box, including low-confidence boxes, and is robust to the brief occlusions common in aerial video. Aerial detection is difficult chiefly because objects are small: a person seen from an altitude of several tens of metres may occupy fewer than fifty pixels, which places the task in the small-object regime where average precision is known to fall most sharply.

Human pose estimation. Two-dimensional pose estimation is dominated by top-down methods, which first detect a person and then regress keypoints within the detected box. Representative models include HRNet [6], which maintains high-resolution representations throughout the network, RTMPose [4], which is optimised for real-time inference, and ViTPose [5], which applies a plain vision transformer. The top-down design is well suited to aerial imagery, in which people are small and sparse, because it allocates the full model resolution to each person rather than sharing it across a largely empty image. We adopt RTMPose as the pose estimator. A recurring limitation of these estimators for aerial use is that their training distributions are dominated by large, ground-level people, which is precisely the distribution shift that RQ1 measures.

Pose-based action and state recognition. Skeleton-based action recognition operates on sequences of estimated keypoints. The spatial-temporal graph convolutional network [7] and its successors treat the skeleton as a graph over time. These methods motivate a pose-based state classifier, although in this work we use a compact feature vector and a multilayer perceptron rather than a graph network, because the number of frames per track is small and the classes are few. The design choice reflects the data regime: a graph convolutional network has many parameters and expects long, reliable skeleton sequences, whereas aerial tracks are short and their keypoints are noisy, so a low-capacity model on engineered features is a more honest baseline.

Search-and-rescue datasets. Datasets that specifically target aerial human understanding include Okutama-Action [8], which provides aerial video with concurrent human action labels, and UAV-Human [9], a large benchmark for human behaviour understanding from drones that includes a pose-estimation subset. The SARD line of work [25] studies person detection in explicit search-and-rescue scenarios. General object context is provided by COCO [10], on which the baseline detector and pose estimator are pretrained.

Image restoration. Classical restoration comprises deconvolution for blur and filtering for noise. The Wiener filter [16] is the linear minimum-mean-squared- error restoration filter, and Richardson-Lucy deconvolution [17, 18] is an iterative method derived from a Bayesian model of image formation. For denoising, the standard spatial methods are the median filter, the bilateral filter [20], and the non-local means filter [19]. Learned deblurring is typically evaluated on the GoPro benchmark [21]; we deliberately restrict our study to the classical regime with a known point spread function so that the measured behaviour reflects the restoration method rather than the boundary handling or an unknown blur.

Local features and matching. The SIFT descriptor [11] and the ORB descriptor [12] are the standard hand-crafted local descriptors, trading accuracy against speed. Learned patch descriptors include L2-Net [13] and HardNet [14], the latter introducing a triplet loss with the hardest negative in the batch. HPatches [15] is the standard benchmark for descriptor evaluation, with viewpoint and illumination sequences and ground-truth homographies. Robust geometric estimation uses RANSAC [22], and the multiple-view geometry underlying homography estimation and mosaicking is standard [23]. Image quality is measured with the peak signal-to-noise ratio and the structural similarity index [24].

3. Data

Table 1 summarises the datasets, their roles, and the scale at which we use them.

Dataset	Role	Scale used
VisDrone-person [1]	detector fine-tuning; restoration frames	548 validation images, 13,969 person boxes
COCO (validation subset) [10]	ground-level pose reference	150 persons, 1,900 keypoints
UAV-Human (pose) [9]	aerial pose evaluation (RQ1)	22,319 persons, 334,217 keypoints
Okutama-Action [8]	state classification; registration; demonstration	multiple videos; the sample video has 2,272 labelled frames
HPatches (sequences) [15]	descriptor matching (RQ4)	116 sequences, 580 image pairs

Table 1. Datasets and the scale at which each is used.

State taxonomy. Okutama-Action provides twelve action labels but no *Waving* class, which we verified directly on the annotations. We therefore group the actions into three SAR-relevant states, defined in `src/config.py`. The mapping is given in Table 2. When a box carries several actions, a precedence rule selects the most urgent state, ordered as motionless, then signalling, then mobile, then stationary, so that the most urgent interpretation dominates. Because Okutama contains no signalling actions, the signalling state does not occur in practice; its presence in the precedence rule documents the intended design. In the sample video the state distribution is 8,433 stationary, 5,559 mobile, 1,337 motionless, and 205 unlabelled boxes.

State	Actions mapped to it
motionless	Lying
stationary	Sitting, Standing, Reading, Drinking, Calling, Hand Shaking, Hugging
mobile	Walking, Running, Carrying, Pushing/Pulling

Table 2. Mapping from Okutama actions to triage states.

The class distribution is strongly imbalanced, with the motionless state, which is the most operationally important, represented by only 1,337 of the labelled boxes. This imbalance is not an artefact to be removed but a property of the deployment setting, in which the people who most need help are the fewest and the hardest to observe, and it is addressed in training by class-balanced loss weighting rather than by resampling.

Restoration degradation protocol. For RQ3 we apply six synthetic degradations with known ground truth to VisDrone frames, defined in `src/eval_restore.py::conditions()`. The conditions are two motion blurs, one defocus blur, two Gaussian-noise levels, and one salt-and-pepper condition, listed in Table 3. Known ground truth permits both intrinsic evaluation (comparing the restored frame against the clean frame) and extrinsic evaluation (measuring the downstream detection and pose accuracy). The professor's specification explicitly permits synthetic degradation for this study.

Condition	Blur	Noise
motion15_n2	linear motion, length 15, angle 45 degrees	Gaussian, sigma 2
motion25_n5	linear motion, length 25, angle 20 degrees	Gaussian, sigma 5
defocus7_n2	disc defocus, radius 7	Gaussian, sigma 2
noise15	none	Gaussian, sigma 15
noise25	none	Gaussian, sigma 25
saltpepper4	none	salt-and-pepper, 4 percent

Table 3. The six degradation conditions.

Matching benchmark. For RQ4 we use the HPatches viewpoint and illumination sequences with their ground-truth homographies. The learned descriptor, TinyDescNet, is trained on self-supervised aerial patch pairs sampled from Okutama frames; these frames do not overlap with the HPatches benchmark, so the descriptor is never trained on the evaluation data.

4. Method

4.1 Pipeline overview

The pipeline comprises six stages: detection, tracking, pose estimation, feature extraction, classification, and the triage overlay. A frame is passed to the detector, which returns person boxes; the tracker links boxes across frames into tracks; the pose estimator returns keypoints for each box; the feature stage converts a short window of a track into a feature vector; the classifier assigns a state; and the overlay draws each person in a colour that encodes the urgency of their state. The stages are deliberately modular, so that a research question can be studied in isolation: RQ1 exercises the pose stage, RQ2 the classification stage, RQ3 a restoration stage inserted before detection, and RQ4 a matching stage that operates on whole frames rather than on individual people.

4.2 Detection and tracking

We use YOLO11s [2] for person detection and ByteTrack [3] for tracking. The detector is fine-tuned on the person-only subset of VisDrone for thirty epochs at an input resolution of 1280 pixels. Fine-tuning adapts the ground-trained model to the aerial domain, in which people are smaller and seen from above than in the COCO pretraining data. The tracker associates detections into tracks using the intersection-over-union of boxes across consecutive frames. ByteTrack retains low-confidence detections for a second association pass, which is valuable in aerial video because a small person frequently drops below the primary confidence threshold for a few frames before reappearing.

4.3 Pose estimation

RTMPose-m [4] is applied top-down to each tracked box, returning seventeen keypoints in the COCO format together with per-keypoint confidences. The top-down design is appropriate for aerial scenes because it processes each person at full model resolution rather than sharing resolution across a sparse image. Each box is expanded slightly and resized to the estimator's input resolution before inference, so that the effective resolution seen by the pose network depends on the original pixel size of the person, which is the mechanism behind the size-dependent accuracy measured in RQ1.

4.4 Feature representation

From the seventeen keypoints and the person box, `src/features.py` constructs a feature vector of dimension 47 for each track window, composed of 34 normalised keypoint coordinates, seven geometric features, and six temporal features.

The keypoints are normalised to be invariant to the person's position and size in the image. Let m_hip be the midpoint of the hips and let the torso length be the distance between the mid-hip and the mid-shoulder. Each keypoint k is transformed as (k minus m_hip) divided by the torso length; when the torso collapses to nearly zero, as in a near-top-down view or when hip and shoulder keypoints are missing, the box diagonal is used as the scale. This yields 34 values (seventeen keypoints, each with two coordinates).

The seven geometric features summarise the posture in one frame. The body-axis angle is the angle from the vertical of the vector from the hips to the shoulders, computed as arctan of the absolute horizontal component over the absolute vertical component; a value near zero indicates an upright person and a value near ninety degrees indicates a lying person. The box aspect ratio, the height divided by the width, complements the angle, because a lying person tends to produce a wide, short box. Four binary flags indicate whether each wrist is above the shoulders or above the nose, which is the signature of a raised arm. The seventh feature is the mean keypoint confidence, which acts as a reliability indicator.

The six temporal features summarise motion over the window, with all distances normalised by the person's own size so that they measure motion relative to the body rather than in raw pixels. The vertical standard deviation of each wrist, relative to the box centre, is large when the arms move up and down, as in waving (two features). The mean speed of each wrist relative to the box centre captures arm motion independently of body motion (two features). The speed of the box centre separates a moving person from a stationary one (one feature), and the standard deviation of the box size captures approach and recession (one feature).

The window length is fifteen frames, and windows overlap so that each track produces several training examples and the per-frame noise is smoothed. The label of a window is the most frequent state among its frames, which is robust to a few mislabelled or missing frames. The temporal features occupy the last six dimensions, which permits a controlled ablation that removes them.

4.5 State classifiers

We compare two classifiers under identical splits and an identical training loop, so that the comparison isolates the representation rather than the training procedure. The PoseMLP is a small multilayer perceptron applied to the 47-dimensional feature vector. The AppearanceCNN is a small convolutional network applied to the raw person crop. Both are trained with the same optimiser, the same number of epochs, and class weights that balance the loss across the unequal state counts; the class weight of a state is the total count divided by the count of that state and by the number of classes. The contrast is deliberate: the PoseMLP sees only the geometry of the skeleton and is blind to appearance, whereas the AppearanceCNN sees pixels and can exploit texture and context but has no explicit notion of posture. Comparing the two under identical conditions is what makes RQ2 a fair test of the value of an explicit pose representation.

4.6 Restoration (RQ3)

We model a degraded frame as $g = h * f + n$, where f is the clean frame, h is a blur point spread function applied by circular convolution, and n is additive noise. Circular convolution is used so that the two-dimensional Fourier transform diagonalises the blur exactly; the measured quality then reflects the restoration method rather than the boundary handling. Writing H for the Fourier transform of the point spread function, G for that of the degraded frame, and the asterisk for the complex conjugate, the methods are as follows.

The pseudo-inverse filter estimates the clean frame as F = H* G / max(|H|^2, epsilon^2). It is retained as an instructive failure case, because wherever |H| is close to zero the division amplifies whatever noise is present.

The Wiener filter is the linear minimum-mean-squared-error restoration filter. For the additive-noise model, its transfer function is W = H* / (|H|^2 + K), where K is a constant that stands in for the ratio of the noise power to the signal power. Where the signal dominates, K is negligible and the filter behaves like the inverse filter; where the noise dominates, K suppresses the corresponding frequencies. We select K from the known noise level and confirm the choice with a sweep.

Richardson-Lucy deconvolution [17, 18] is an iterative method that begins from a flat estimate and, at each iteration, multiplies the current estimate by the back-projected ratio of the observed frame to the current prediction. Because it enforces non-negativity and is derived from a Poisson noise model, it preserves edges better than the linear filters, at a substantially higher cost. We run twenty iterations.

The spatial baselines are the Gaussian, median, bilateral [20], and non-local means [19] filters, together with unsharp masking as a sharpening baseline. These methods do not require the point spread function and therefore apply to any degradation, but they cannot invert a blur. The median filter is nonlinear and is particularly suited to impulsive salt-and-pepper noise, whereas the bilateral and non-local means filters preserve edges by averaging only over photometrically similar neighbourhoods.

4.7 Matching and registration (RQ4)

With a fixed SIFT keypoint detector, we compare three descriptors: the SIFT descriptor [11], the ORB descriptor [12], and TinyDescNet, a compact convolutional descriptor trained with the HardNet loss [14] on self-supervised aerial patch pairs. Descriptors are matched with the Lowe ratio test, which accepts a match only when the nearest-neighbour distance is smaller than 0.8 times the second-nearest distance. Fixing the detector isolates the descriptor as the single variable under study, so that any difference in matching accuracy is attributable to the description of the patch rather than to where the patches are placed.

For the applied result, we estimate a homography between frames with RANSAC [22]. The number of RANSAC iterations required to obtain, with probability p, at least one sample free of outliers is N = log(1 minus p) / log(1 minus w^s), where w is the inlier fraction and s is the minimal sample size, which is four for a homography. We chain the homographies to a common reference frame, build a mosaic, and project the foot point of each track onto the mosaic under a planar-scene assumption, which produces a single victim map.

4.8 Evaluation metrics

All metrics are our own implementations in `src/`, so that no result depends on a library whose internal choices we cannot inspect.

Detection. We use the average precision at an intersection-over-union threshold of 0.5, denoted AP50. The intersection-over-union of a predicted box A and a ground-truth box B is IoU(A, B) = area(A intersect B) / area(A union B). A prediction is a true positive when its intersection-over-union with an unmatched ground-truth box is at least 0.5, and a false positive otherwise; unmatched ground-truth boxes are false negatives. With precision = TP / (TP + FP) and recall = TP / (TP + FN) computed as the confidence threshold is swept, AP50 is the area under the resulting precision-recall curve.

Pose. We use the percentage of correct keypoints at a threshold of 0.1, denoted PCK@0.1. A predicted keypoint p is correct when its Euclidean distance to the ground-truth keypoint g satisfies the norm of (p minus g) at most 0.1 times a reference length d taken from the person box, and PCK is the fraction of correct keypoints, PCK = (1 / K) times the sum over the K keypoints of the correctness indicator. The score is reported both overall and stratified by person pixel height, which is the stratification that exposes the size effect in RQ1.

Classification. We use the macro-averaged F1 score, macro-F1 = (1 / C) times the sum over the C classes of F1_c, where F1_c = 2 P_c R_c / (P_c + R_c) is the harmonic mean of the per-class precision and recall. Because the classes are averaged without weighting, a rare class such as motionless contributes as much as a common one, which is the appropriate emphasis for triage.

Restoration. We use the peak signal-to-noise ratio and the structural similarity index [24]. For an eight-bit image the peak signal-to-noise ratio is PSNR = 10 log10 (255^2 / MSE), where MSE is the mean squared error between the restored and the clean frame; a higher value indicates a smaller pixel error. The structural similarity index between local windows x and y is SSIM = (2 mu_x mu_y + C1)(2 sigma_xy + C2) / ((mu_x^2 + mu_y^2 + C1)(sigma_x^2 + sigma_y^2 + C2)), where mu, sigma, and sigma_xy are the local means, variances, and covariance, and C1 and C2 are small stabilising constants; it rewards structural rather than purely pixel-wise agreement. We also report the wall-clock runtime per frame.

Matching. We use the mean matching accuracy at a pixel threshold t, denoted MMA at t, which is the fraction of proposed matches whose reprojection error under the ground-truth homography is below t pixels, averaged over the pairs. We also report the mean homography corner error, the mean distance between the four image corners mapped by the estimated homography and by the ground-truth homography, which summarises the geometric quality of the estimate in a single number of pixels.

5. Experiments and Results

All full-scale numbers use the fine-tuned detector checkpoint `models/yolo11s_visdrone_best.pt`, and all evaluators are our own implementations in `src/`. The complete tables are stored in `results/`.

5.1 Detection foundation

Detection is the foundation of the pipeline, so we first quantify the aerial domain gap for detection. On the 548-image VisDrone-person validation set, which contains 13,969 person boxes, the zero-shot COCO model reaches an AP50 of 0.437, and fine-tuning for thirty epochs raises it to 0.709 and the recall to 0.859 (Table 4). The fine-tuning curves are shown in Figure 1.

detector	AP50	recall
YOLO11s zero-shot (COCO)	0.437	0.596
YOLO11s fine-tuned (30 epochs)	**0.709**	**0.859**

Table 4. Detection on VisDrone-person validation.

The increase of 0.27 in AP50 confirms that the aerial domain gap for detection is substantial and that a modest fine-tune closes much of it.

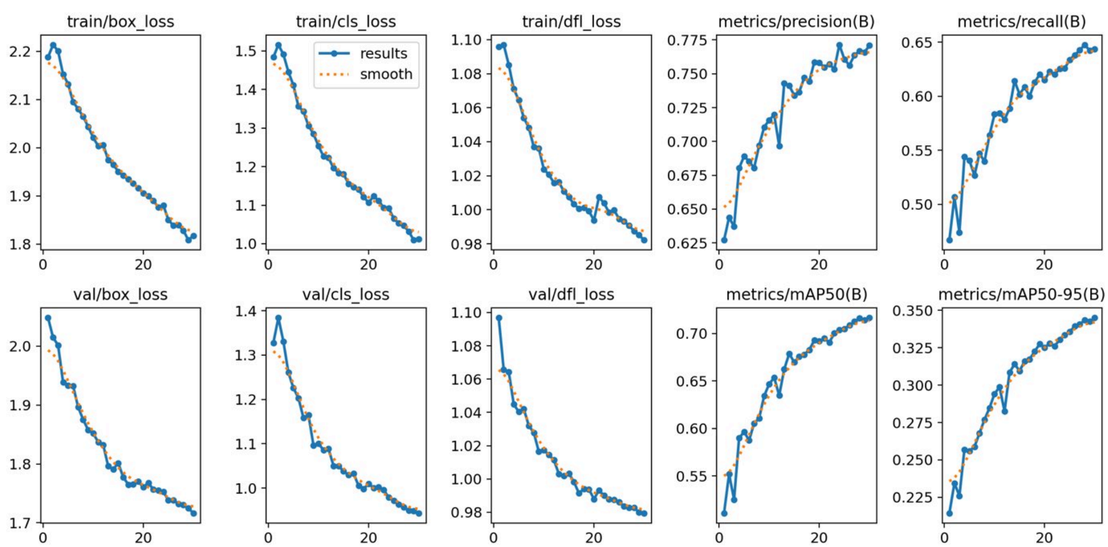

Figure 1. Fine-tuning curves for the VisDrone-person detector (notebook 01). The box, classification, and distribution-focal losses decline smoothly on both the training and validation splits, and the precision, recall, and mean-average- precision curves rise and plateau, which indicates stable convergence without overfitting over the thirty epochs.

The confusion matrix of the fine-tuned detector, in Figure 2, shows that the residual error is dominated by the background axis: the model misses a fraction of small persons and raises a small number of background detections, rather than confusing people with any other class, which is consistent with a single-class small-object detection problem.

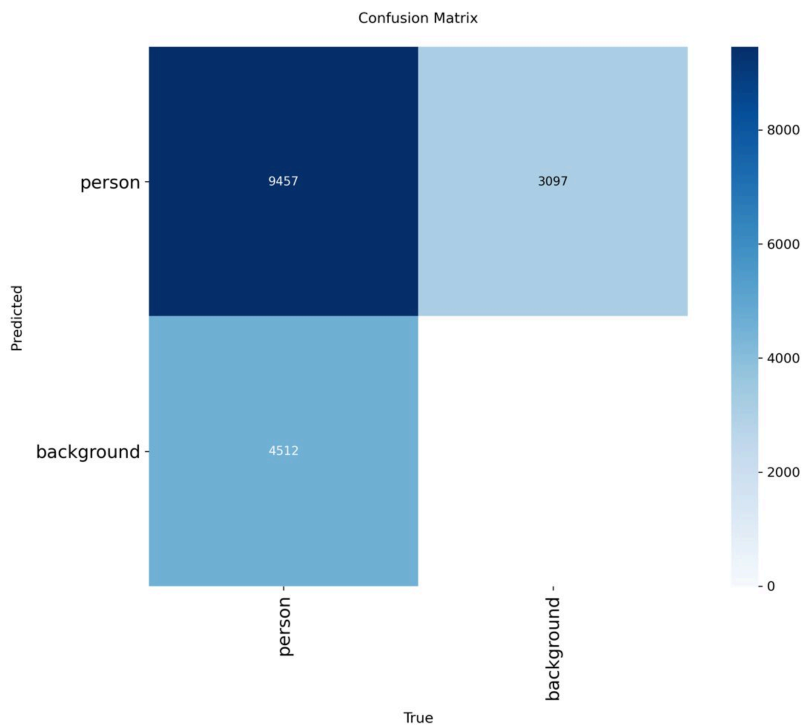

Figure 2. Confusion matrix of the fine-tuned detector on the validation set; the residual error is between the person class and the background, not a class confusion.

5.2 RQ1 — the aerial pose gap is a function of person size

The ground-level reference measurement, on a COCO subset of 150 persons, gives a PCK@0.1 of 0.95 and already shows the size effect, with 0.958 above one hundred pixels against 0.90 in the fifty-to-one-hundred-pixel band. This validates the estimator and the evaluator. We then apply RTMPose-m zero-shot over all UAV-Human frames, evaluating 22,319 persons and 334,217 keypoints. Table 5 gives the result stratified by person height, and Figure 3 plots it.

domain	PCK@0.1	25–50 px	50–100 px	100+ px
ground-level (COCO)	0.95	–	0.90	0.958
aerial (UAV-Human)	0.942	**0.091**	**0.440**	0.944

Table 5. Pose accuracy on ground-level and aerial data, stratified by height.

The overall aerial PCK is within one point of the ground-level value, which is misleading, because when the accuracy is stratified by person height it declines sharply: it remains high above one hundred pixels, is approximately halved in the fifty-to-one-hundred-pixel band, and approaches total failure below fifty pixels. The aerial pose gap is therefore primarily a function of person size rather than of viewpoint. A downscale ablation, in which the persons are reduced to thirty-five percent of their size before pose estimation, reproduces the same degradation, which isolates scale as the cause. This finding also explains the RQ2 result below, because the pose features become unreliable precisely where the keypoints do.

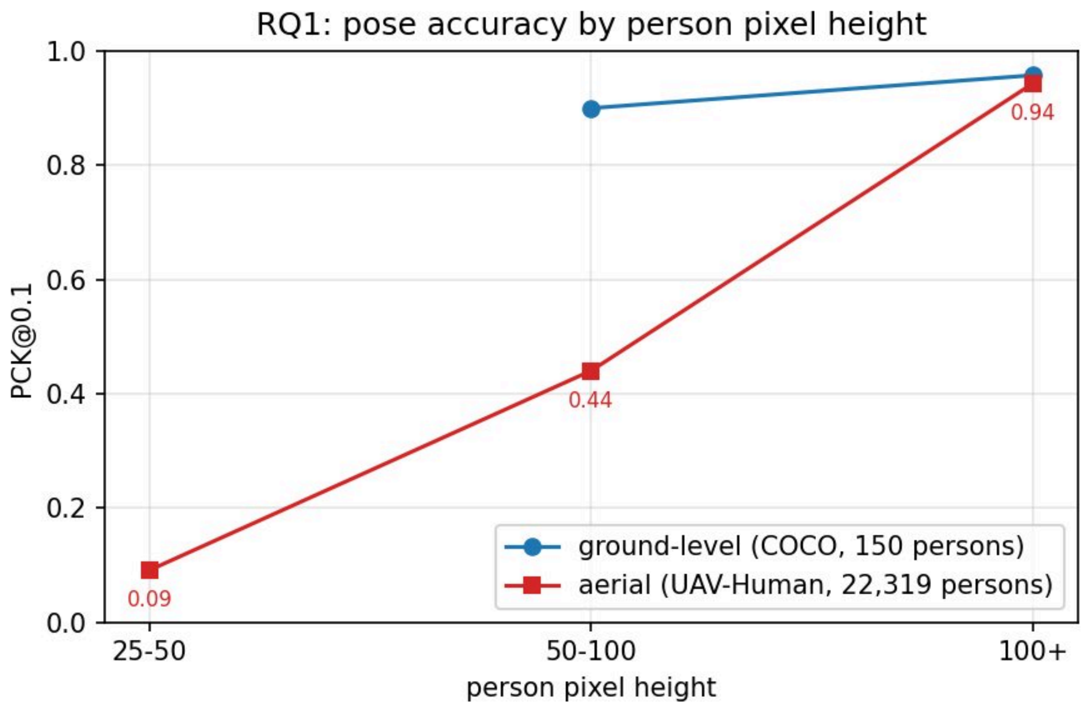

Figure 3. PCK@0.1 against person pixel height: ground-level against aerial. The curves coincide for large persons and diverge steeply below one hundred pixels, which is the visual statement of the size-driven gap.

As a label-free recovery attempt, we train a student pose model, yolo11n-pose, on Okutama pseudo-labels for twenty epochs, reaching a pose mean average precision at an intersection-over-union of 0.5 of 0.508 and a mean average precision averaged over thresholds of 0.307. Figure 4 shows the student's training curves, which converge stably; a full before-and-after PCK quantification of this recovery is left to future work (Section 9).

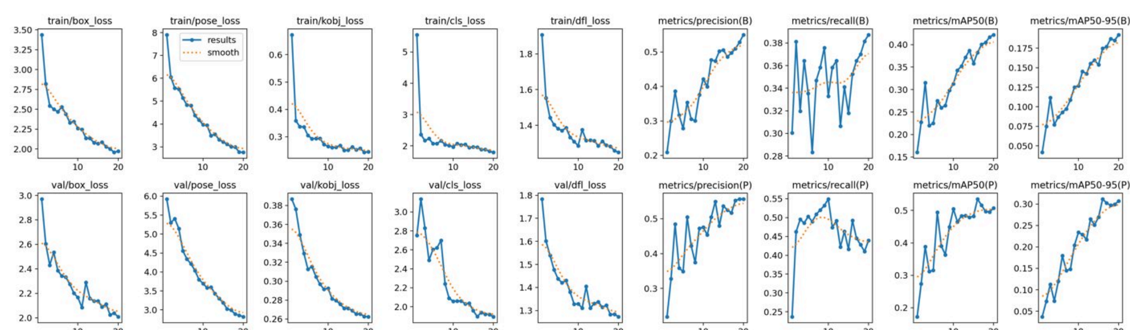

Figure 4. Training curves of the yolo11n-pose student fine-tuned on Okutama pseudo-labels; the pose and box losses decline and the pose mean average precision rises over the twenty epochs.

5.3 RQ2 — pose against appearance

We hold out whole videos, which is stricter than a track-level split and tests generalisation across scenes. Both models use identical splits and training. Table 6 gives the result, and Figures 5 and 6 give the confusion matrices.

model	macro-F1	F1 mobile	F1 stationary	F1 motionless
PoseMLP (pose features)	0.363	0.631	0.458	0.000
AppearanceCNN (raw crops)	**0.399**	0.611	0.588	0.000

Table 6. State classification under a video-level split.

Two findings emerge. First, both models fail on the rare motionless class, with an F1 of zero, whereas at single-video scale the same models reached between 0.5 and 0.7; video-level generalisation is considerably harder, and the motionless class, which corresponds to persons lying down, is the most affected. Second, the appearance model narrowly outperforms the pose model, but the margin is only 0.036, which is consistent with RQ1: the pose features are limited by unreliable keypoints on persons between thirty and eighty pixels tall, while the appearance model loses the track-level background cue it could otherwise exploit once whole videos are held out.

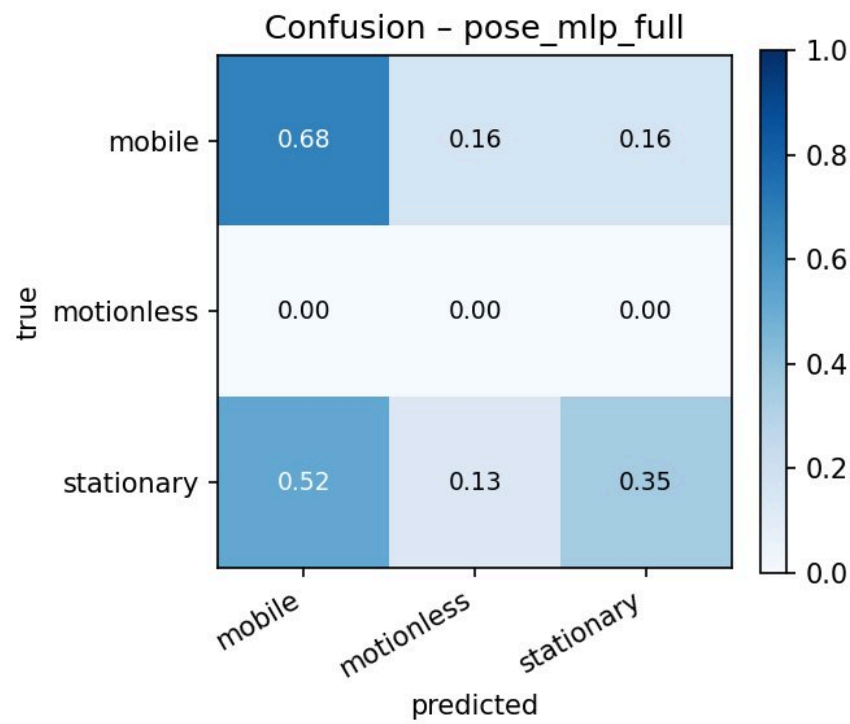

Figure 5. Confusion matrix of the PoseMLP classifier under the video-level split; the motionless row is empty, which is the zero-F1 failure reported in Table 6.

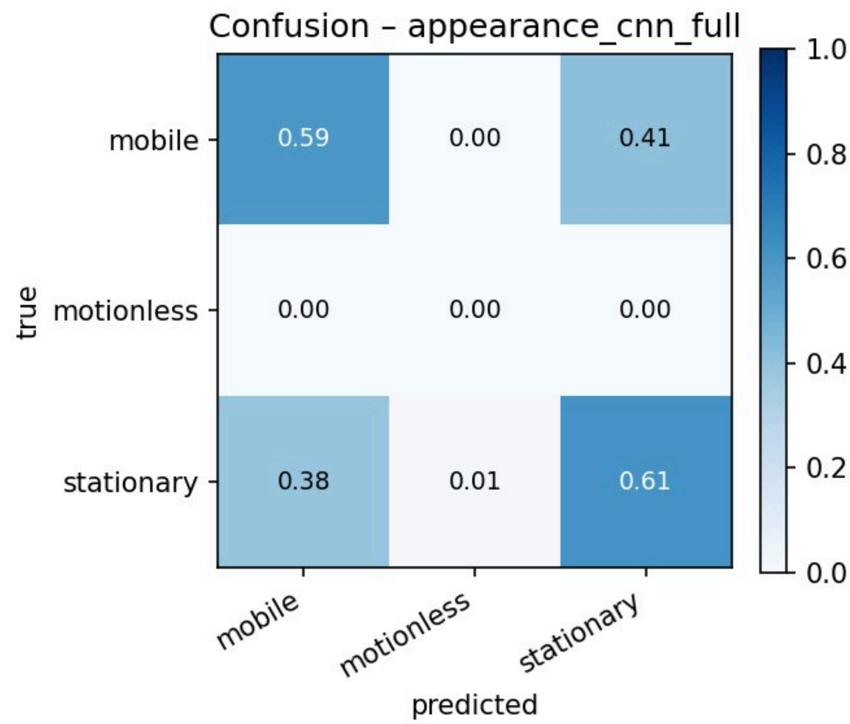

Figure 6. Confusion matrix of the AppearanceCNN classifier under the same split; the mobile and stationary classes are separated but the motionless class is again never recovered.

5.4 RQ3 — restoration

Intrinsic quality. Table 7 gives the peak signal-to-noise ratio, in decibels, for every method and condition, over two hundred frames. The best method in each condition is shown in bold.

condition	none	inverse	wiener	rl20	unsharp	gaussian	median	bilateral	nlm	butterworth
motion15_n2	23.8	8.4	25.9	**26.7**	23.3	23.8	–	–	–	–
motion25_n5	22.5	6.1	23.5	**24.1**	21.4	22.8	–	–	–	–
defocus7_n2	23.3	6.4	**25.5**	25.2	23.1	23.4	–	–	–	–
noise15	24.7	–	–	–	–	28.1	28.1	**30.8**	29.3	26.7
noise25	20.5	–	–	–	–	25.7	25.7	**27.3**	26.7	26.1
saltpepper4	19.1	–	–	–	–	26.0	**27.8**	19.1	21.2	25.8

Table 7. Intrinsic restoration quality (PSNR, dB) over 200 frames.

For the blur conditions, the frequency-domain methods perform best. Richardson- Lucy is marginally the strongest on motion blur and the Wiener filter is strongest on defocus, but the Wiener filter matches Richardson-Lucy to within a few tenths of a decibel at a fraction of the runtime: the Wiener filter runs in about ninety milliseconds per frame, whereas Richardson-Lucy with twenty iterations runs in about three seconds. The pseudo-inverse filter exhibits the expected catastrophic noise amplification, falling to between six and eight decibels. For the noise conditions, the nonlinear spatial filters perform best: the bilateral filter is strongest on Gaussian noise and the median filter is strongest on salt-and-pepper noise, where the bilateral filter in fact degrades the image below the noisy baseline. Each noise type therefore requires a different filter. The Wiener K sweep peaks at a value of 0.046, shown in Figure 7, and the qualitative comparisons in Figures 8 and 9 illustrate the same conclusions on representative frames.

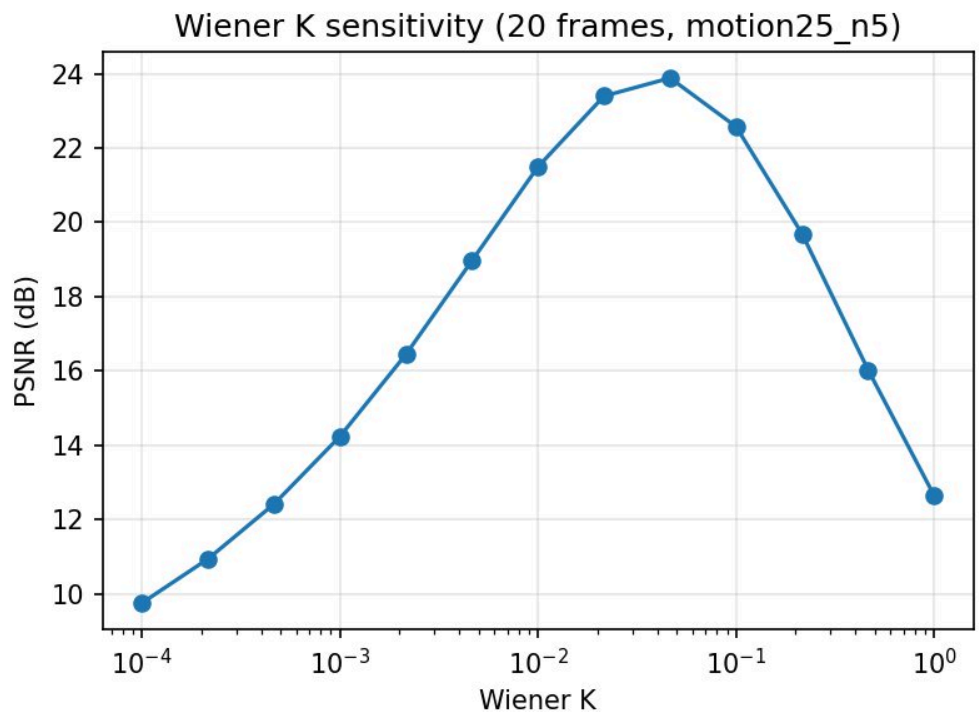

Figure 7. Wiener-filter K sensitivity (PSNR against K); the curve peaks near K = 0.046, the value used throughout.

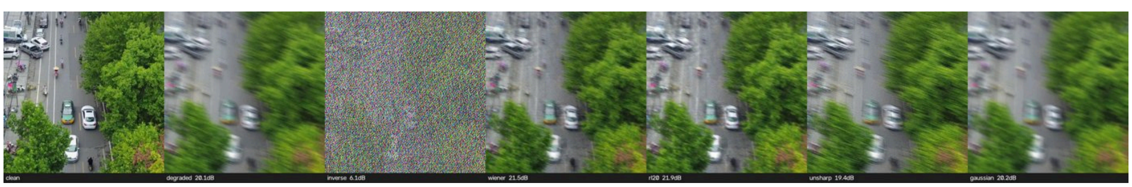

Figure 8. Qualitative restoration under the motion25_n5 condition: the degraded frame, the Wiener and Richardson-Lucy deconvolutions, and the spatial baselines. The deconvolutions recover edges that the spatial filters leave blurred.

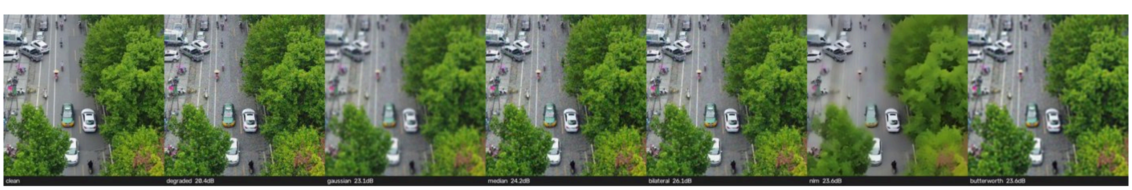

Figure 9. Qualitative restoration under the noise25 condition. The bilateral and non-local means filters give the smoothest result, but as the extrinsic evaluation shows, this smoothing removes the small persons that detection depends on.

Extrinsic evaluation. The central question of the section is whether restoration recovers task performance. Table 8 gives the detection AP50 over all 548 images, with the fine-tuned detector.

condition	clean	degraded	wiener	rl20	unsharp	median	nlm	butterworth
motion25_n5	0.709	0.025	**0.091**	0.079	0.011	–	–	–
noise25	0.709	0.460	–	–	–	**0.496**	0.387	0.428

Table 8. Detection AP50 under degradation and restoration, 548 images.

The principal finding holds at full scale: an improvement in pixel quality does not imply an improvement in task performance. On noise, the non-local means filter attains a high peak signal-to-noise ratio (26.7 decibels in Table 7) yet reduces detection accuracy to 0.387, below the degraded baseline of 0.460, by smoothing away small persons; the median filter is the only method that improves detection, to 0.496. At blur comparable to the person size, detection accuracy is not recoverable, falling to 0.025 and reaching only 0.091 after the Wiener filter, because the information is destroyed rather than merely obscured.

A complementary pose evaluation at local scale, over two hundred UAV-Human persons, shows the opposite, milder trend for blur, because pose is measured on larger, already-detected persons. Under motion blur the clean PCK of 0.958 falls to 0.931 when degraded and is restored to 0.952 by the Wiener filter; under noise it falls to 0.937 and is restored to 0.948 by the median filter. Table 9 gives the detail. The overall lesson is that a restoration method must be validated on the downstream task, and that the appropriate method depends on which task is downstream.

condition	variant	PCK	PCK 50–100 px	PCK 100+ px
motion15_n2	clean	0.958	0.500	0.960
motion15_n2	degraded	0.931	0.250	0.933
motion15_n2	wiener	0.952	0.667	0.953
noise25	degraded	0.937	0.583	0.938
noise25	median	0.948	0.583	0.949

Table 9. Pose PCK under degradation and restoration, 200 persons.

5.5 RQ4 — matching and the victim map

Table 10 gives the matching results over all 116 HPatches sequences, comprising 285 illumination pairs and 295 viewpoint pairs. The metrics are the mean matching accuracy at one, three, five, and ten pixels, the mean homography corner error in pixels, and the runtime per pair in milliseconds.

method	condition	MMA@1	MMA@3	MMA@5	MMA@10	corner err (px)	ms/pair
SIFT	illumination	0.522	0.700	0.729	0.747	31.6	148
SIFT	viewpoint	0.466	0.710	0.747	0.766	51.7	251
ORB	illumination	0.434	0.675	0.732	0.760	46.4	52
ORB	viewpoint	0.283	0.670	0.736	0.759	119.9	79
TinyDescNet	illumination	0.505	0.646	0.669	0.684	33.9	2047
TinyDescNet	viewpoint	0.383	0.601	0.638	0.659	93.5	2320

Table 10. Descriptor matching on HPatches, all 116 sequences.

SIFT attains the highest accuracy and the lowest homography corner error, particularly under viewpoint change. ORB exchanges accuracy for a three-to-five- fold speedup and degrades sharply at the strict one-pixel threshold under viewpoint change, where its mean matching accuracy falls to 0.283 against 0.466 for SIFT; this is adequate for coarse alignment but weaker for building a map. The self-supervised TinyDescNet nearly matches SIFT under illumination change, with a mean matching accuracy at one pixel of 0.505 against 0.522, but performs worse under viewpoint change, which is consistent with its training on aerial appearance variation rather than on geometric warps. TinyDescNet is slow here because it is evaluated on the central processing unit; on a graphics processing unit it is considerably faster. Figures 10 and 11 plot the accuracy against the pixel threshold and the precision against recall, and make the ordering visible across all thresholds rather than at a single operating point.

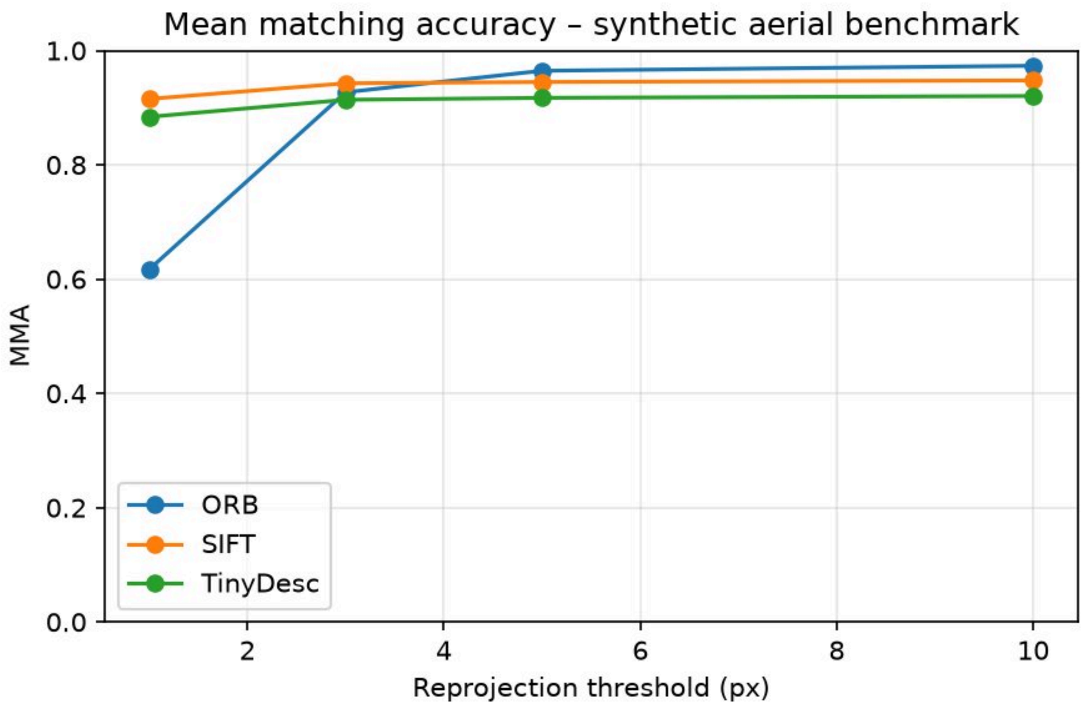

Figure 10. Mean matching accuracy against the pixel threshold for the three descriptors, separated by illumination and viewpoint change.

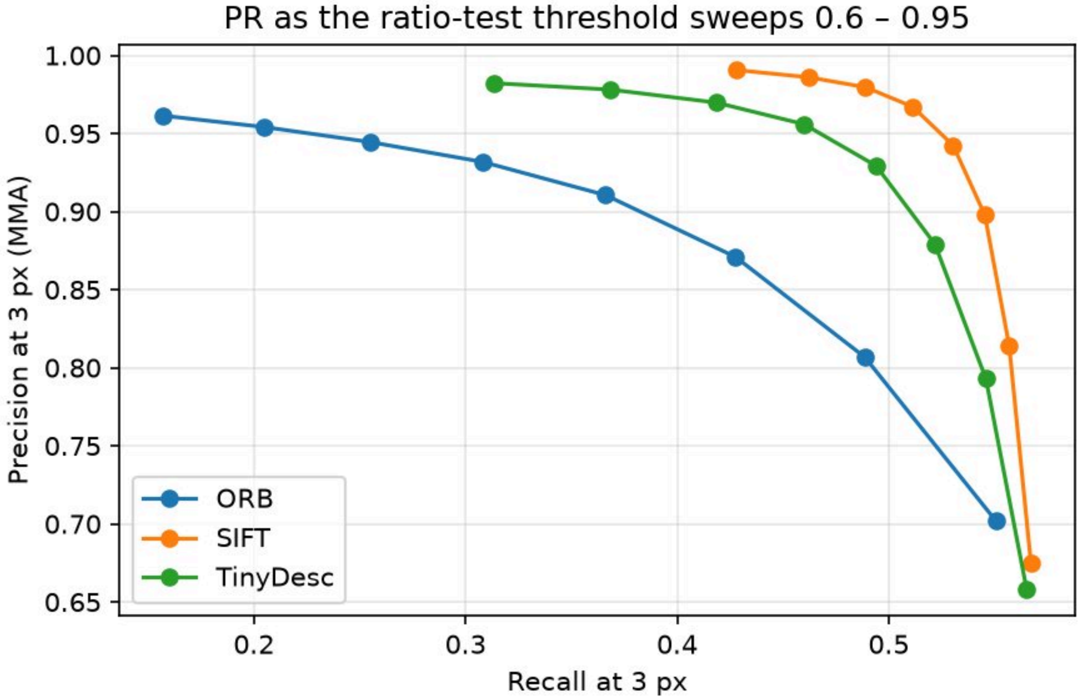

Figure 11. Precision against recall for the three descriptors; SIFT dominates, with TinyDescNet close under illumination change and ORB fastest but least precise.

Applied result. Using SIFT and RANSAC, we register every twelfth frame of a 288-frame Okutama sweep, obtaining a mean of about nine hundred and five inliers per link, with a minimum of six hundred and ten. We chain the homographies to the first frame and build a mosaic of 2200 by 1104 pixels (Figure 12). Projecting the foot point of each pipeline track onto the mosaic produces a single victim map (Figure 13), which converts the per-frame triage overlay into one operator-facing map of the sweep.

Figure 12. SIFT and RANSAC mosaic of the 288-frame Okutama sweep.

Figure 13. Victim map: per-track states projected onto the mosaic, coloured by triage urgency.

5.6 Error analysis

The most instructive failures are consistent with the quantitative results across the four questions.

- **Small persons (RQ1, RQ2).** The dominant failure mode is missed detections and unreliable pose on persons below fifty pixels. This is the direct cause of the zero F1 on the motionless class, because a person lying down at low resolution produces a small, low-confidence set of keypoints from which the body-axis feature cannot be computed reliably.
- **Occlusion and clutter (RQ2).** Persons partially occluded by objects or by one another produce truncated pose estimates, which the classifier tends to assign to the stationary class by default.
- **Pose collapse on lying persons.** For a person lying down, the projected skeleton is short and wide, and the mid-hip to mid-shoulder axis is poorly defined, which reduces the discriminative power of the body-axis angle, the feature intended to separate motionless from the other states.
- **Noise amplification (RQ3).** The pseudo-inverse filter amplifies noise into visually unusable frames, as reflected in its peak signal-to-noise ratio of six to eight decibels; this is the expected behaviour of the inverse filter where the transfer function is near zero, and it motivates the regularised Wiener filter.
- **Mosaic drift (RQ4).** Because the homographies are chained frame to frame, the registration error accumulates over a long sweep, and the mosaic drifts. Loop closure or bundle adjustment would reduce this drift.

6. Demonstration

The demonstration video, `results/videos/demo_final.mp4`, runs the full pipeline — the fine-tuned detector, ByteTrack, RTMPose, the PoseMLP state classifier, and the triage overlay — over 1,800 frames of a held-out Okutama video, with each person coloured by the urgency of their state. The video was rendered with the canonical detector, so that it is consistent with the quantitative results. Figure 14 shows a representative frame.

Figure 14. A representative frame from the demonstration: each tracked person carries an identifier, an estimated skeleton, and a state label, and is coloured by triage urgency.

7. Discussion

The four questions are separate experiments, but their answers reinforce a single account of the aerial triage problem, which this section draws together.

Person size is the governing variable. The most consistent theme across the results is that the difficulty of aerial understanding is set by the pixel size of the person and not by the aerial viewpoint as such. RQ1 shows this directly for pose, with accuracy collapsing below fifty pixels; the detection confusion matrix in RQ1's foundation shows the same effect as missed small persons; and RQ2 inherits it as the zero-F1 motionless class. The practical implication is a resolution budget: for triage to work, each person must be observed at a sufficient pixel size, which constrains the flight altitude, the sensor resolution, and the ground sampling distance jointly. A pipeline that is accurate in principle can still fail in the field simply because the drone flew too high.

When an explicit pose representation would help. RQ2 finds that an explicit pose representation does not yet beat an appearance baseline under a strict video-level split. This should not be read as evidence that pose is unhelpful in general, but that its advantage is conditional on reliable keypoints, which the small-person regime denies. Where persons are larger — a lower flight, a higher-resolution sensor, or a zoomed sweep of a candidate location — the pose features would be expected to recover the advantage that the RQ1 analysis predicts, because the body-axis and wrist-motion features that separate the states are computable only when the skeleton is reliable.

Restoration must be chosen for the task, not for the pixel metric. RQ3 establishes that peak signal-to-noise ratio is a misleading objective for a detection pipeline: the non-local means filter improves the pixel metric while lowering detection accuracy, because it removes exactly the small, low-contrast structures that detection depends on, whereas the median filter, which is less impressive on the pixel metric, improves the task. The operational recommendation that follows is to select and tune a restoration stage against the downstream detector, and to prefer a mild, structure-preserving filter over an aggressive denoiser. Motion blur at the scale of the person is a separate warning: once the blur kernel is comparable to the person size, the information is destroyed and no restoration recovers the task, so blur is best prevented at capture by a faster shutter or by gimbal stabilisation.

Descriptor and mapping choices for deployment. RQ4 gives a clear ordering for the mapping component. SIFT is the right default when accuracy governs, ORB is the right choice when a limited onboard compute budget governs and coarse alignment suffices, and a small learned descriptor is competitive only within its training regime. For the victim map itself, the chained-homography mosaic is adequate over a short sweep but drifts over a long one, so a deployed system would add loop closure or bundle adjustment and would georeference the mosaic using the drone's position and camera intrinsics, converting the map from mosaic coordinates into coordinates a ground team can act on.

Taken together, the discussion converts the four measurements into design guidance: fly low enough to resolve people, restore for the detector rather than for the eye, prefer pose only where keypoints are reliable, and close the map's drift before trusting it operationally.

8. Conclusions

This work built a complete aerial search-and-rescue triage pipeline and used it to answer four questions with consistent, full-scale evidence. The aerial pose gap is primarily a function of person size rather than of viewpoint: accuracy is close to the ground-level value for large persons but falls to 0.440 in the fifty-to-one-hundred-pixel band and to 0.091 below fifty pixels. For triage-state classification under a strict video-level split, an appearance baseline slightly outperforms an explicit pose representation, with macro-averaged F1 scores of 0.399 against 0.363, and both models fail on the rare motionless class, which locates the difficulty in the small-person regime identified in RQ1. For restoration, an improvement in pixel quality does not imply an improvement in task performance: on additive noise, the non-local means filter raises the peak signal-to-noise ratio yet lowers detection accuracy, whereas the median filter improves it, so a restoration method must be validated on the downstream task. For matching, SIFT remains the most accurate and geometrically precise descriptor, ORB is substantially faster at a cost in precision, and a compact self-supervised descriptor approaches SIFT under illumination change but not under viewpoint change. Taken together, these results give a realistic account of where an aerial search-and-rescue pipeline succeeds and where it fails.

9. Assumptions, Limitations, and Future Work

Assumptions. The state taxonomy assumes the three search-and-rescue-relevant classes, because Okutama-Action provides no signalling class. Pose accuracy is measured with PCK@0.1 normalised by the person box, which assumes that the box is an adequate proxy for scale at aerial resolution. The restoration study assumes a known point spread function and models blur with circular convolution. The registration and victim map assume a planar scene, so that a homography is an adequate frame-to-frame model, and the foot-point projection assumes that people stand on flat ground.

Limitations and future work.

- **No signalling class.** Okutama-Action contains no *Waving* class, so the taxonomy is motionless, stationary, and mobile; a signalling class would require a dataset such as UAV-Gesture.

- **RQ1 recovery.** The pseudo-label student model is trained, but its before-and-after PCK recovery is not yet quantified at full scale; this and the UAV-Human fine-tuning variant are the natural next step.
- **RQ2 ablations.** The window-size, crop-resolution, and temporal-feature ablations described in notebook 03 are not included at full scale; they, and a detection resolution comparison between 640 and 1280 pixels, remain to be run.
- **RQ3 setting.** The setting is non-blind, in that the point spread function is known, and uses circular convolution, so there are no realistic boundary effects; blind point-spread-function estimation and learned deblurring are future work, and the gap to a real-blur benchmark such as GoPro [21] would quantify the cost of these simplifications.
- **RQ4 mapping.** The registration assumes a planar scene and drifts over long sweeps; loop closure or bundle adjustment would reduce this drift, and the victim map is in mosaic coordinates rather than geographic coordinates, which would require the camera position and intrinsics.
- **Scale of some artefacts.** The registration and victim map are demonstrated on one video, and the pose-under-degradation evaluation is at local scale.

Appendix A — Reproducibility and Implementation

Each module of the project is a single notebook, `notebooks/01` through `notebooks/06`, and the README lists the exact commands. The full-scale detection and pose training ran on Colab graphics processing units; the remaining experiments ran locally on an Apple M1 Pro, using the Metal Performance Shaders backend and the central processing unit. Random seeds are fixed in the scripts.

All evaluators are our own implementations in `src/`: `eval_detect.py` for average precision, `eval_pose.py` for the percentage of correct keypoints, `eval_restore.py` for the peak signal-to-noise ratio, the structural similarity index, and the degradation suite, and `eval_match.py` for the matching metrics. The pipeline stages are implemented in `detect.py`, `track.py`, `pose.py`, `features.py`, `classify.py`, `restore.py`, `match.py`, `register.py`, and `render_demo.py`, with shared configuration in `config.py`. The result tables and figures are in `results/`, and the numerical results are also collected in `results/RESULTS.md`.

Table A1 lists the principal settings, so that the experiments can be reproduced.

Component	Setting
Detector	YOLO11s, input 1280 px, 30 epochs, VisDrone-person subset
Student pose model	yolo11n-pose, 20 epochs, Okutama pseudo-labels
Pose estimator	RTMPose-m, top-down, 17 COCO keypoints
Feature window	15 frames, overlapping, majority-vote label
Feature vector	47 dimensions (34 keypoint, 7 geometric, 6 temporal)
Classifier loss	class-balanced weighting across the three states
Wiener regularisation K	0.046, selected by sweep
Richardson-Lucy	20 iterations
Lowe ratio test	0.8
RANSAC (homography)	minimal sample 4, success probability target 0.99
Registration stride	every 12th frame of the sweep

Table A1. Principal hyperparameters and settings.

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